

EE8216 ADVANCED COMPUTER NETWORKS PROJECT PROPOSAL

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Project Topic: Mobility Models and the Effect of Mobility on Wireless Network Performance – Artificial Intelligence Focus

Project Definition: In dynamic ad hoc mobility networks such as Mobile Ad Hoc Networks (MANETs) and Vehicular Networks (VANETs), topologies constantly change. Traditional fixed-network protocols fail in these environments because they cannot adapt to the rapid breaking and re-forming of links. Although dynamically adaptive, MANETs and VANETs rely heavily on reactive protocols. This project explores the potential transition toward cognitive networking – a theoretical framework where Artificial Intelligence (AI) and Machine Learning (ML) are used to predict mobility patterns and proactively heal network paths while concurrently adjusting security protocols.

Problem Statement: Mobile networks face challenges such as: 1) Volatility: Unpredictable traffic caused by the dynamic nature of nodes and devices can overload network capacity in crowded areas. 2) Overhead: Standard route repair consumes excessive bandwidth. 3) Masking: High-mobility environments are subjective to vulnerabilities and malicious attacks by intruders. [1]

Background and Significance: The motivation behind this project proposal is the growing need to shift away from the break-and-fix mindset toward a predictive and proactive model to prevent communication blackouts and network instabilities as topologies evolve, consumer demands change, and data traffic volume increases. For instance, as we move toward 6G, the Internet of Vehicles (IoV) demands sub-millisecond latency and autonomous systems such as drones and self-driving cars require continuous link reliability. The significance of this study lies in its predictive, context-aware framework and approach, ensuring dynamic networks can leverage AI and ML to be more resilient and high-performance in the modern mobile networking era.

Project Objectives:

- **Investigate Mobility Models and Network Contextual Awareness:** To explore different mobility models and networks such as VANETs and MANETs and understand their functions, how various mobility patterns (Urban, Highway, Random) work and can integrate with intelligent algorithms.
- **Analyze Predictive Mitigation:** To understand common mobile network challenges. To analyze predictive modeling options and algorithms in anticipating link failures (fault detection), conducting root cause analysis, initiating self-healing routines, predicting traffic patterns, and promoting energy efficiency and a green network, among other potential use cases.
- **Understand AI-Enabled Security Mechanisms:** To examine how network security mechanisms can dynamically adjust their sensitivity and protocols based on the observed mobility context to balance protection and performance using AI and ML.
- **Apply to Real-World Cases:** To understand and apply the framework and its methods to real-world examples and discuss how results can be measured against Key Performance Indicators (KPIs).

References: [1] Anderson, Thomas E., et al. *The Case for a Reduced Instruction Set Computer*. Technical Report UW-CSE-93-11-03, University of Washington, 1993, dada.cs.washington.edu/research/tr/1993/11/UW-CSE-93-11-03.pdf. Accessed 30 Jan. 2026.